

Introduction

The database is the essential form for understanding the logics of digital information systems; Through the layers of platform assemblages there exists a materiality, a substance. But can data be managed in the same way as a record? Does the management of records differ from the management of information? What might LIS professionally contribute to a diverse information ecology. For my project I looked at a particular organization called The Syllabus, examining how they harvest, store, and curate data for their clients. The goal was to efficiently indexalize their database records to improve access and the speed of data retrieval. Yet, the Syllabus database lacked standardization, making for a fascinating yet painstaking database design. A diversity of sources with differing data qualities had produced hard to reconcile data structures. The absence of an appropriate method (incomplete normalization) for accommodating and validating a high variety of data endorsed a variety of data quality issues: language field inconsistencies, malformed author arrays, and ad-hoc field names betrayed any chance at a meaningful interoperability layer. Under this rubric indexicalization was but sheepskin wrapped around a wolf. Through the steps of this project though, it became glaringly more apparent that what was necessary was a descriptive schema designed to facilitate consistent representation across the stages of the pipeline. Through the re-engineering of tables into categories, an environment emerges: all data is its metadata and all metadata is its architecture. The following reflection walks through some of my thought process across the entirety of the project and within the minutia of the classification system and the descriptive schema.

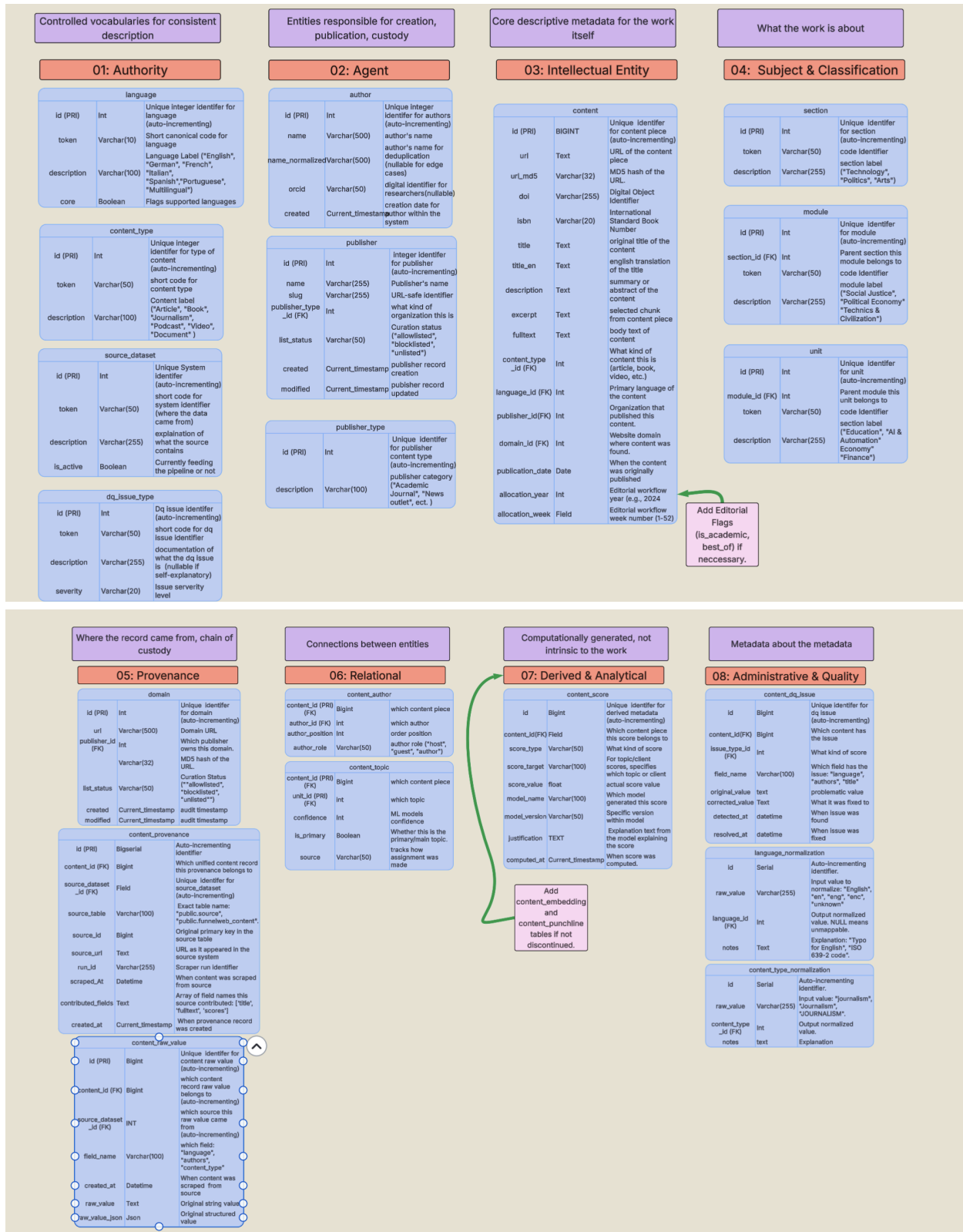
STFC system

The STFC system defines the characteristics of The Syllabus's source across time and space. It is modeled on what users might be looking for when they seek out a source from the Syllabus. In a sense it is quite skeleton-boned but I have not found a reason to change it, as the objective is to deliver content sources to users. The Syllabus is a recommendation engine in that sense – less about the author of a journal or the imposition of particular research, but content. So the delivery of STFC necessitates this mission. The Facet layers are essentially de-materialized lego blocks that can be arranged and constructed in whichever order, but they are always and only a subtle mixture of the underlying components and their characteristics. Within them there may exist a tree structured index that allows us to efficiently locate all data entries with search key values in a desired range. The node or facet at which you start at depends upon the nature of your search. As you move down this chain, you essentially have a subtree that contains the data you are after. This system of course has its flaws, one being that all facets are not equal. It is much harder to query by subject than it is by language or content type, for example. It is my understanding that Raganathan's facet and colon classification became ontologically more precise over time. With more time, I would like to dissolve the confusions I have between data science principles of information management and more classical understandings of record-keeping within our profession. For example, could a fully expressive notation that represents very complex subjects evolve with primary key identifiers? Regardless, at the moment it feels as if there is some flexibility, or maybe generalizability, to facet classification that allows for expressive power within the context of The Syllabus.

Descriptive Schema

The trials and tribulations of indexicalizing messy data brought me a new descriptive schema, and ultimately a controlled vocabulary for interoperability between The Syllabus's existing

tables. AS I worked on the project more and more, it became about reducing the complexity of queries through indexicalization. To do that, the data needed to be broken down into further tables under the laws of different classes or ontologies. Then that structure needed to reflect both the underlying medium—a relational knowledge repository—and the STFC system. Encouraged by PREMIS, I first identified the core Intellectual Entity or content piece and its corresponding fields. This is the core manifestation of the work so to speak. This core table subsequently produces junction tables, or a relational network of connections that engenders a resource but maintains metadata about the metadata. Controlled authority fields evoke consistent description in a one-to-many encoding—there are many resources or intellectual entities but all of them are only ever represented by one language or one content type. Agent fields are those responsible for creation, publication and custody while Subject & Classification deals with what the work is about. Awaiting both author and topic is a many-to-many relationship with a content piece: a work can have multiple authors and multiple authors can work on different pieces, just as a topic can be represented by a variety of sources, so to can a source be represented by a variety of topics. The relational category below in **figure two** builds towards such complexity. Moreover, Curation planning and management of the data encompasses administrative quality control (data stewardship), derived metadata (computational affordances), and provenance (Data Lineage and transformation history), all of which makes more a functionally more sound schema, improving the efficiency and speed of indexical queries while reducing complexity of retrieval overall.



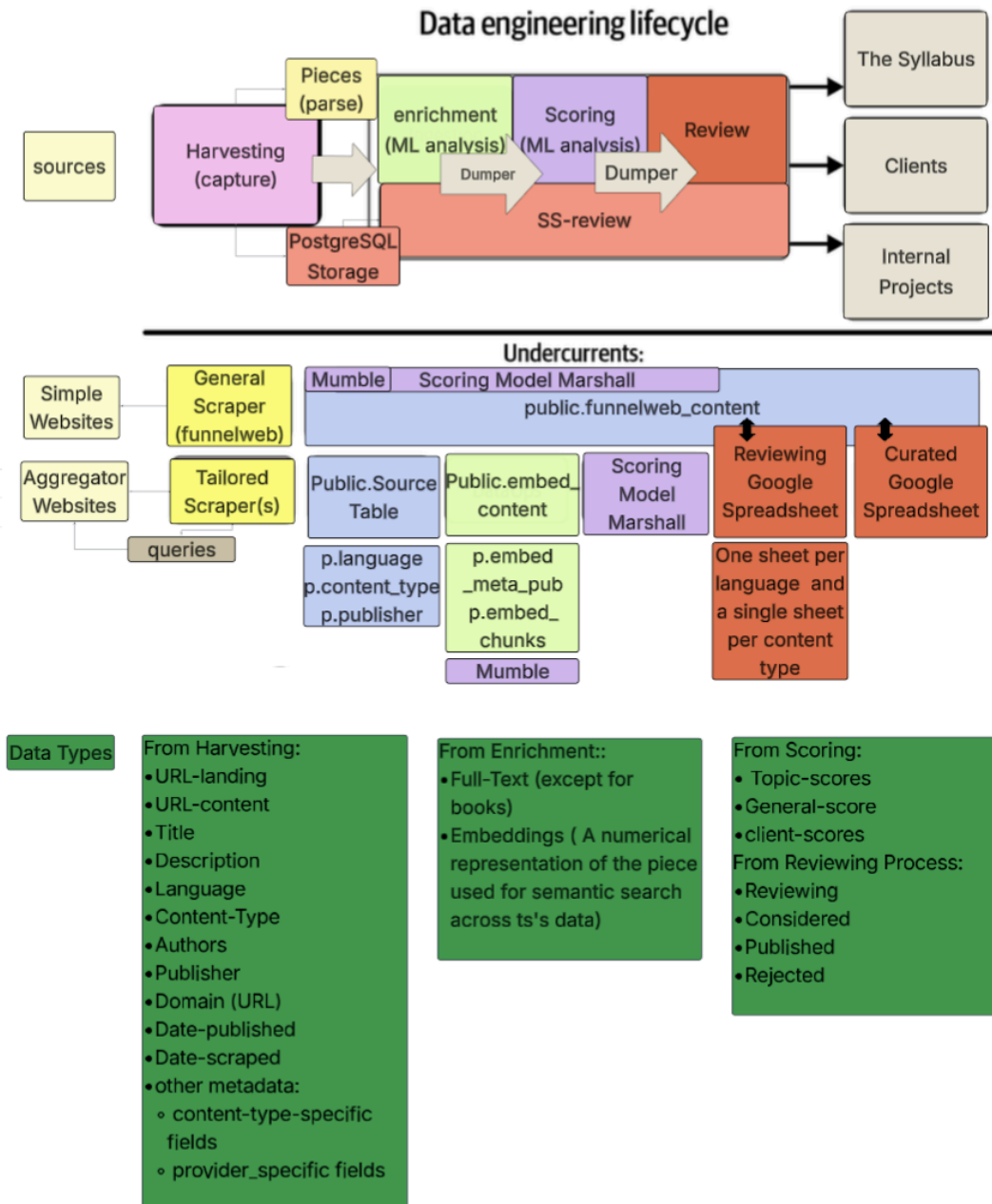
Figures 1 & 2: Descriptive Metadata Schema

Conclusion

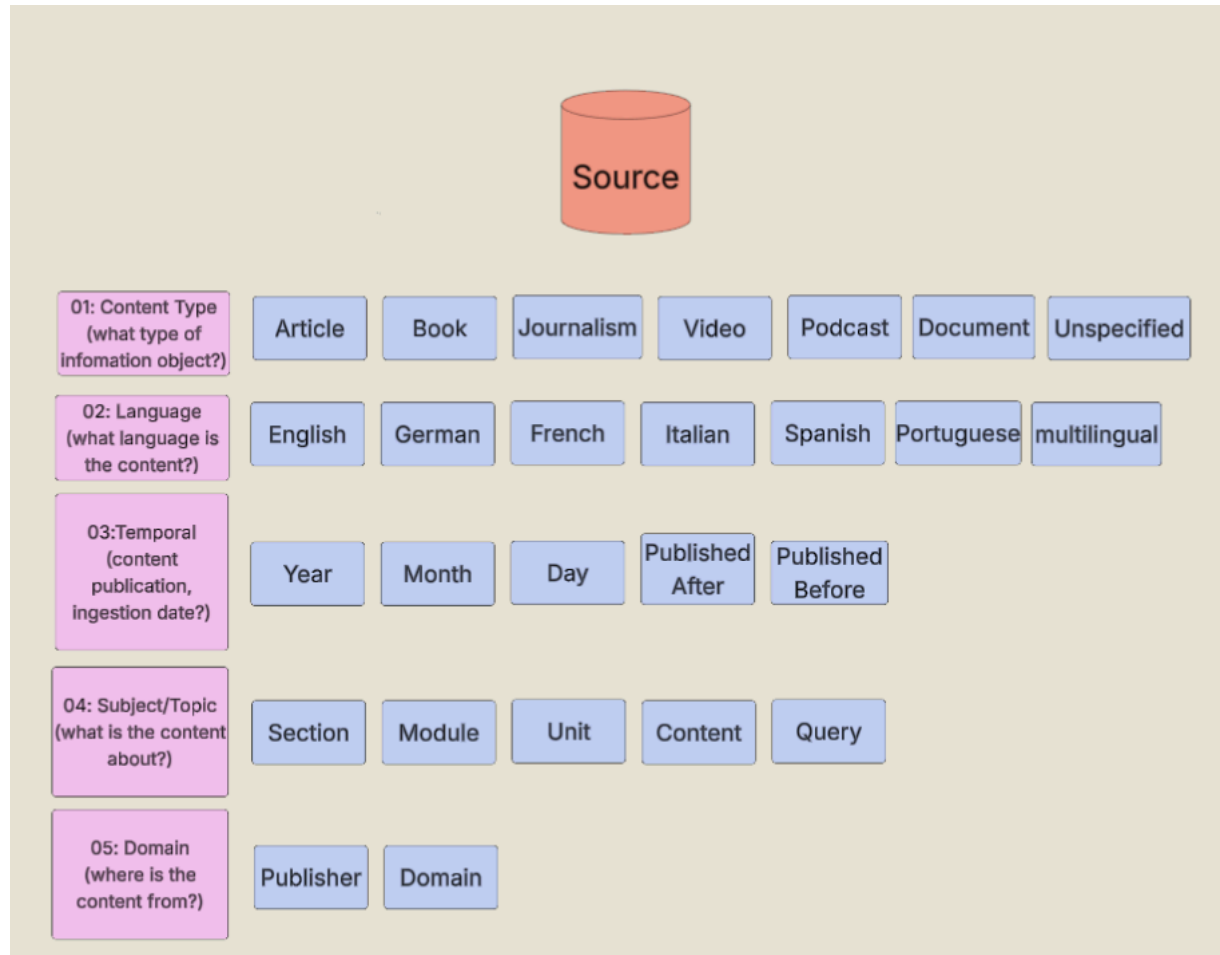
All data is local, inseparable from its source system. Proper functionality and efficiency within a database is not only dependent upon quality data, but the curational planning process overall. Data engineering lifecycle models seek to explain, but what's missing from their exploratory power is the altogether different lifecycle of digital curation; that is, the managerial, administrative, collaborative, and developmental documentation that helps to determine and prepare the right methodologies or frameworks for the collection or creation of data is altogether more important than the actions of the lifecycle. Ingestion is tied to migration, and so it goes onward through the digital curation landscape.

Appendix

Appendix C - The Syllabus's Data Engineering Lifecycle =



Appendix B - Source Type Facet Classification (STFC)



Appendix C - Descriptive Resource Examples

Resource 1:

Original Attribute	Target Attribute	Target Value
Title	<i>Dc: title</i>	"Zohran Mamdani: "Hope Is Alive""
<i>content_type_id → content_type.name</i>	<i>dc:type</i>	"Journalism" "public.content_type_id/3"
<i>language_id → language.description, language.token</i>	<i>dc: language</i>	"English" "public.language_id/1" language.token: en
<i>publisher_id → publisher.name, publisher.slug</i>	<i>dc:publisher</i>	"The Nation" "publisher.id/13693"
url, url_md5, PK id	<i>dc:identifier</i>	url_md5: "urn:md5:600dfbd510259ae72b2e5df6533393ed" PK id: "public.source/22851675" url: https://www.thenation.com/article/politics/zohran-mamdani-victory-speech/
publication_date	<i>dc:date</i>	created: "2025-11-05"

Resource 2:

Original Attribute	Target Attribute	Target Value
<i>Title</i>	<i>Dc: title</i>	“Generative AI and misinformation: a scoping review of the role of generative AI in the generation, detection, mitigation, and impact of misinformation ”
<i>content_type_id</i> → <i>content_type.name</i>	<i>dc:type</i>	“Article” “public.content_type_id/1”
<i>language_id</i> → <i>language.description</i> , <i>language.token</i>	<i>dc: language</i>	"English" “public.language_id/1” language.token: en
<i>publisher.id</i> → <i>publisher.name</i> ,	<i>dc:publisher</i>	“AI & Society” “publisher.id/18018”
<i>url</i> , <i>url_md5</i> , PK id	<i>dc:identifier</i>	PK id: "public.source/23947419" url: https://link.springer.com/article/10.1007/s00146-025-02620-3
<i>publication_date</i>	<i>dc:date</i>	created: "2025-09-30"

Resource 3:

Original Attribute	Target Attribute	Target Value
<i>Title</i>	<i>Dc: title</i>	“Destrucción de palestina es la destrucción de la tierra”
<i>content_type_id</i> → <i>content_type.name</i>	<i>dc:type</i>	“Book” “public.content_type_id/2”
<i>language_id</i> → <i>language.description</i> , <i>language.token</i>	<i>dc: language</i>	"Spanish" “public.language_id/5” language.token: es
<i>publisher_id</i> → <i>publisher.name</i> ,	<i>dc:publisher</i>	“Versos Libros” “publisher.id./18018”
<i>url</i> , <i>url_md5</i> , PK id	<i>dc:identifier</i>	PK id: "public.source/22625407" url: https://www.todostuslibros.com/libros/destruccion-de-palestina-es-la-destruccion-de-la-tierra_978-84-10344-91-4
<i>publication_date</i>	<i>dc:date</i>	created: "2025-09-01"

Resource 4:

Original Attribute	Target Attribute	Target Value
<i>Title</i>	<i>Dc: title</i>	“Brazil and the World System w/ Fernando Haddad”
<i>content_type_id</i> → <i>content_type.name</i>	<i>dc:type</i>	“Podcast” “public.content_type_id/5”
<i>language_id</i> → <i>language.description</i> , <i>language.token</i>	<i>dc: language</i>	"English" “public.language_id/1” language.token: en
<i>publisher_id</i> → <i>publisher.name</i> ,	<i>dc:publisher</i>	“The Dig” “publisher.id/13640”
<i>url</i> , <i>url_md5</i> , PK id	<i>dc:identifier</i>	PK id: "public.source/23904365" url: https://thedig.blubrry.net/podcast/brazil-and-the-world-system-w-fernando-haddad
<i>publication_date</i>	<i>dc:date</i>	created: "2025-11-04"

Appendix D: Descriptive Schema Architecture

